Shivashriganesh P. Mahato

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BIOGRAPHY

I am a third-year PhD student at the University of Pennsylvania, advised by Prof. Eva L. Dyer, interested in representation learning from neural data. My work spans invasive (intracranial EEG, neural spikes, calcium imaging) and non-invasive (EEG, EMG) neural modalities as I develop new approaches for learning how to decode from brain signals. Of particular interest to me is the challenge of integrating methods across sources of heterogeneity, including temporal drift, subject-to-subject variability, and diverse behaviors. I am interested in developing flexible tools that can enable clinical outcomes downstream, empower assistive technologies, and hopefully enhance our understanding of the brain.

EDUCATION

PhD in Computer and Information Science — Aug 2024 – Present
University of Pennsylvania (transferred from Georgia Tech)
Advisor: Prof. Eva L. Dyer

MS in Computer Science — Aug 2022 – Aug 2023
Texas A&M University
Advisor: Prof. Yoonsuck Choe • Thesis: *Comparison of Evolved Memory Mechanisms in Neural Networks*

BS in Applied Mathematics — Aug 2019 – May 2022
Texas A&M University
Minors in Computer Science and Statistics • Summa Cum Laude with Honors

PUBLICATIONS

** denotes co-first authors*

Population-Aware Contrastive Channel Embeddings for Intracranial EEG
Mahato, S.P.*, Balaji, K.*, Lachi, D., Dyer, E.L.
Machine Learning for Health (ML4H) 2026

BrainWideBench: Benchmarking Large-Scale Pretraining and Across-Animal Transfer in Multi-Region Neural Recordings
Andre, A.*, **Mahato, S.P.***, Arora, V., Balaji, K., Lachi, D., Krishna, N.H., Xiao, J., Zhang, Y., Mao, X., Ma, W., Yu, H., International Brain Laboratory, . . . , Winter, O., Richards, B., Lajoie, G., Hurwitz, C., Azabou, M., Whiteway, M.R., Paninski, L., Dyer, E.L.
arXiv preprint arXiv:2609.22064 [paper] 2026

Scaling Neural Motor Decoding via Decoupled Behavioral Pretraining
Lachi, D., Arora, V., **Mahato, S.P.**, Andre, A., Dyer, E.L.
Neural Information Processing Systems (NeurIPS) 2026

Zero-Shot Across-Subject Neural Decoding via Neuron-Level Representations
Arora, V., Lachi, D., **Mahato, S.P.**, Andre, A., Poschl, J., Chapuis, G., Whiteway, M.R., Dyer, E.L.
NeurIPS Workshop on Symmetry and Geometry in Neural Representations (NeurReps) 2026

A Scalable Self-Supervised Method for Modeling Human Intracranial Recordings during Natural Behavior
Mahato, S.P.*, Xiao, J.*, Andre, A., Chau, G., Ma, W., Knight, I.J., Nguyen, D., Hu, L.J., Brunton, B.W., Beauchamp, M.S., Pesaran, B., Shuvaev, S.A., Dyer, E.L.
NeurIPS Workshop on Foundation Models for the Brain and Body **Spotlight (top 15%)** [paper] 2025

Exploiting All Laplacian Eigenvectors for Node Classification with Graph Transformers
Arora, V.*, Lachi, D.*, **Mahato, S.P.**, Azabou, M., Chen, Z., Dyer, E.L.
NeurIPS Workshop on New Perspectives in Graph Machine Learning [paper] 2025

Learning from Multi-Table Relational Data with the Relational Graph Perceiver
Lachi, D., Mohammadi, M., Meyer, J., Arora, V., **Mahato, S.P.**, Palczewski, T., Dyer, E.L.
EurIPS Workshop on AI for Tabular Data [paper] 2025

An Open-Source Ecosystem for Models of Multi-Modal Brain and Body Data
Azabou, M., Arora, V., Sobral, M., Suarez, L., Krishna, N., Ryoo, A., Mao, X., **Mahato, S.P.**, Andre, A., Paninski, L., Lajoie, G., Richards, B., Dyer, E.L.
NeurIPS Workshop on Foundation Models for the Brain and Body, Demo Track 2025

Use of External Markers by Reactive Agents as an Easier Evolutionary Route Toward Memory
Mahato, S.P., Kaliyur, S., Chung, J.R., Choe, Y.
International Joint Conference on Neural Networks (IJCNN) [paper] 2024

RESEARCH EXPERIENCE

Graduate Research Assistant (PhD), Neural Data Science (NerDS) Lab Aug 2024 – Present
PI: Prof. Eva L. Dyer • University of Pennsylvania

- Co-led BrainWideBench, a benchmark for neuro-foundation models on the IBL Brain-wide Map spanning decoding, neural activity prediction, and anatomical localization; established classical and foundation-model baselines in collaboration with the International Brain Laboratory (under review, ICLR 2027).
- Developed a self-supervised framework for human intracranial EEG that learns general representations by reconstructing masked channels (NeurIPS 2025 workshop, spotlight), and population-aware contrastive channel embeddings for iEEG (under review, ICASSP 2027).
- Contributed to large-scale and zero-shot cross-subject neural decoding (NeurIPS 2026; NeurReps 2026) and to graph learning with spectral positional encodings and relational data (NeurIPS and EurIPS 2025 workshops).
- Core contributor to `torch_brain`, an open-source library for deep learning on neural data (NeurIPS 2025 workshop, demo track).

Graduate Researcher (MS Thesis), Brain Networks Lab Jan 2023 – Aug 2023
PI: Prof. Yoonsuck Choe • Texas A&M University

- Used neuroevolution (NEAT) to compare memory-capable controllers on a memory-dependent RL task, showing that feedforward agents leaving external markers in their environment act as a rudimentary memory mechanism and offer an easier evolutionary route to memory than recurrent architectures (IJCNN 2024).

PROFESSIONAL EXPERIENCE

Senior Associate Software Engineer, Capital One Aug 2023 – Aug 2024
Chicago, IL

- Led design, implementation, and full-stack testing (E2E, performance, resiliency) of a major payments scheduling feature; set team-wide performance testing standards as SME.
- Built an automated payments reconciliation reporting service and improved observability of payments infrastructure through end-to-end tracing, metrics, and logging.

Software Engineer II, Sellers Shield Jan 2023 – Mar 2023
Austin, TX

- Standardized backend testing infrastructure and led OAuth implementation for the partner-facing API.

Software Engineering Intern, Capital One Jun 2022 – Aug 2022
Plano, TX

- Built a serverless AWS application for audit log validation during platform migration, cutting validation from days to hours.

Software Engineering Intern, Ascension Digital Studio May 2021 – Aug 2021
Austin, TX

- Shipped features for a surgical scheduling system on GCP and cut test time by ∼45% by embedding MongoDB in the test suite.

Contract Software Developer, Vitalus Health Feb 2021 – May 2021
Katy, TX

- Built an SMS API, patient scheduling system, and data provisioning layer supporting customer communication and chatbot integration.

TEACHING EXPERIENCE

Teaching Assistant, CIS 7000: Deep Learning for Time Series (DL4TS) Spring 2026
University of Pennsylvania

Teaching Assistant, Transformers in Neuroscience Tutorial Mar 2025
Computational and Systems Neuroscience (COSYNE) • Montreal, Canada

Teaching Assistant, ECE 4252/8803: Fundamentals of Machine Learning Fall 2024
Georgia Institute of Technology

CONFERENCES & WORKSHOPS

NeurIPS 2026 Dec 2026 (upcoming)
Sydney, Australia
Poster (co-presenter): *Scaling Neural Motor Decoding via Decoupled Behavioral Pretraining*
Poster (co-presenter): *Zero-Shot Across-Subject Neural Decoding via Neuron-Level Representations*

ARNI Annual Retreat Sep 2026
New York, NY
Talk: *Measuring Progress in Neuro Foundation Models with BrainWideBench*

NeurIPS 2025 Dec 2025
San Diego, CA
Oral & Poster: *A Scalable Self-Supervised Method for Modeling Human Intracranial Recordings*
Poster (co-presenter): *Exploiting All Laplacian Eigenvectors for Node Classification with Graph Transformers*

CCN 2025 Workshop Sep 2025
Functional Alignment of Information Encoded in Neural Activity • Dartmouth, Hanover, NH

Adaptive Deep Brain Stimulation Workshop Apr 2025
University of Pennsylvania, Philadelphia, PA
Talk: *Machine Learning for Adaptive DBS*

Data-Driven and Large-Scale Modeling in Neuroscience Workshop Apr 2025
Georgia Tech, Atlanta, GA
Talk: *Large-Scale Deep Learning in Neuroscience*

COSYNE 2025 Mar 2025
Montreal, Canada
Teaching Assistant: *Transformers in Neuroscience Tutorial*

IEEE WCCI 2024 Jun 2024
Yokohama, Japan
Oral: *Use of External Markers by Reactive Agents as an Easier Evolutionary Route Toward Memory*

TALKS

Measuring Progress in Neuro Foundation Models with BrainWideBench Oct 2026
International Brain Laboratory (IBL) Lab Meeting

BrainWideBench: Benchmarking Large-Scale Pretraining on Multi-Region Neural Recordings Feb 2026
ARNI Working Group on Neuro-Foundation Models • New York, NY

Large-Scale Deep Learning in Neuroscience Jun 2025
Computational Neural-Engineering Training Program Tutorial Series • Georgia Tech, Atlanta, GA

HONORS, AWARDS & CERTIFICATIONS

- Top Reviewer, NeurIPS (2026)
- President's Fellowship, Georgia Institute of Technology (2024)
- IEEE Computational Intelligence Society (CIS) "Industry Early Career" Participation and Travel Grant (2024)
- AWS Certified Solutions Architect, Associate (May 2024 – May 2027)
- Honors Degree in Mathematics, Texas A&M Department of Mathematics (2022)
- Certificate of Excellence, Texas A&M Department of Mathematics (2022)
- Classroom Excellence Award for MATH 470 (Cryptography), Texas A&M Department of Mathematics (2021)

PROFESSIONAL AFFILIATIONS

- Reviewer, Neural Information Processing Systems (*NeurIPS*) (2026)
- Reviewer, Computational and Systems Neuroscience (*COSYNE*) (2026, 2027)
- Reviewer, NeurIPS 2025 Workshop on Foundation Models for the Brain and Body (2025, 2026)
- Reviewer, International Joint Conference on Neural Networks (*IJCNN*) (2024, 2025)
- Reviewer, IEEE Transactions on Cognitive and Developmental Systems (2024, 2025)
- Member, Institute of Electrical and Electronics Engineers (2024 – 2025)
- Member, IEEE Computational Intelligence Society (2024 – 2025)

MENTORSHIP

- Keshav Balaji, MS Computer and Information Science, University of Pennsylvania (Ongoing)
- Reid Laughton, MS Computer and Information Science, University of Pennsylvania (2026)
- Jake Manning, MS Computer Science, Texas A&M University (2025)
- Tanu Shree, MS Computer Science, Texas A&M University (2023)
  - Thesis: *Classification of Cricket Songs Based on Their Genus and Species*

TECHNICAL SKILLS

- **ML Tools:** pytorch, lightning, numpy, cupy, pandas, sklearn, keras, neat-python, wandb, slurm
- **Languages:** Python, Java, C++, Javascript, R, SQL
- **Web Tools:** Spring Boot, Django, NodeJS, Flask, React, VueJS
- **Dev Tools:** Git, Docker, Kubernetes, Linux, Jenkins, Splunk, New Relic, Jira, Terraform
- **Cloud:** AWS, Google Cloud Platform (GCP), Firebase, Cloudflare